

Generative AI Core Concepts and Basics

BSIT DATA SCIENCE LECTURE HANDOUT

Models • Prompts • LLMs • Diffusion • Responsible Use

Contents

1	Introduction to Generative AI	2
1.1	Where Generative AI Fits	2
1.2	Why Generative AI Matters	2
2	Generative AI vs. Traditional Predictive AI	3
3	Core Building Blocks	3
3.1	Tokens and Context	3
3.2	Embeddings	4
4	How Generative AI Works	4
4.1	Learning from Data	4
4.2	Training, Fine-Tuning, and Inference	4
4.3	Text Generation Intuition	4
5	Main Families of Generative Models	4
5.1	Large Language Models	5
5.2	Diffusion Models	5
5.3	GANs and VAEs	5
6	Transformers and Attention	5
7	Prompting Basics	6
7.1	What Makes a Good Prompt?	6
7.2	Common Prompting Styles	6
8	Retrieval, Grounding, and RAG	6
9	Applications of Generative AI	7
9.1	Examples Relevant to BSIT Data Science	7

10 Limitations and Risks	7
10.1 Hallucination	8
10.2 Prompt Injection and Safety	8
11 Evaluating Generative AI Systems	8
12 Best Practices for Students and Developers	8
13 Mini Case Study for BSIT	9
14 A Small API-Style Example	9
15 Summary	9
16 Review Questions	10

1 Introduction to Generative AI

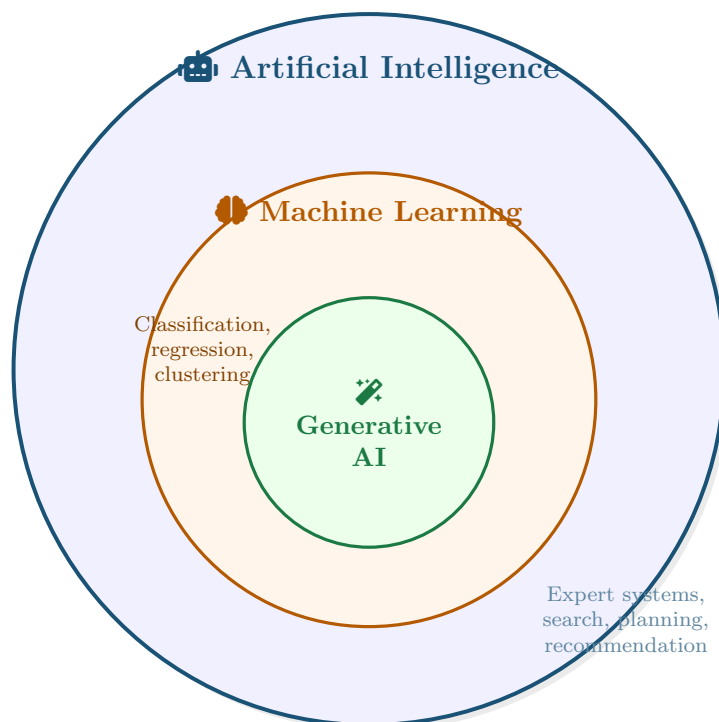
Generative AI

Generative AI is a branch of artificial intelligence that can **create new content** such as text, images, audio, code, video, and synthetic data. Instead of only classifying or predicting labels, it learns patterns from existing data and produces new outputs that resemble that data.

Simple Meaning

If a model can write a paragraph, create an image from a prompt, complete computer code, summarize a document, or generate a voice clip, it is acting as a **generative** model.

1.1 Where Generative AI Fits



- **AI** is the broad field of making machines behave intelligently.
- **Machine Learning** learns patterns from data.
- **Generative AI** focuses on creating new outputs based on learned patterns.

1.2 Why Generative AI Matters

- It helps people create content faster.
- It supports chatbots, digital assistants, coding tools, design tools, and educational tools.
- It is widely used in text generation, image generation, translation, summarization, and content personalization.
- It is transforming how humans interact with software.

2 Generative AI vs. Traditional Predictive AI

Aspect	Predictive / Discriminative AI	Generative AI
Main goal	Predict labels or values	Create new content
Typical output	Class, score, forecast	Text, image, code, audio, video
Common question	“What is this?”	“Create something like this”
Examples	Spam detection, house price prediction	ChatGPT, image generators, code assistants

💡 Easy Distinction

If a model tells you whether an email is spam, that is predictive AI. If a model writes a new email for you, that is generative AI.

3 Core Building Blocks

Concept	Meaning
Prompt	The instruction or input given to the model
Token	A small unit of text such as a word, part of a word, or symbol
Embedding	A numeric representation that captures meaning or similarity
Model	The trained system that generates outputs
Parameter	A learned internal value of the model
Inference	The stage where the model produces output for a new input
Context window	The amount of input the model can consider at one time
Temperature	A setting that affects randomness and creativity of the output
Fine-tuning	Additional training to specialize a general model for a task or domain
Multimodal model	A model that can work with more than one data type such as text and images

3.1 Tokens and Context

📄 Token

A token is a small piece of text processed by the model. A sentence is not read as one whole unit. It is split into smaller parts, and the model works through them step by step.

💡 Context Window

The context window is like the model's working memory for the current interaction. If important information is outside that window, the model may ignore it or forget it.

3.2 Embeddings

📖 Embedding

An embedding is a vector representation of data that places similar items close together in a mathematical space. In practice, this helps models compare meaning, search documents, recommend related items, and support retrieval systems.

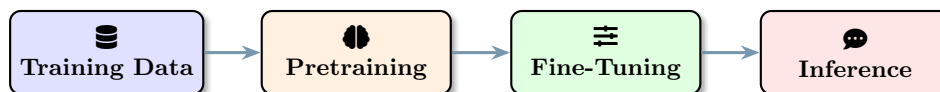
4 How Generative AI Works

4.1 Learning from Data

💡 General Idea

Generative models are trained on large collections of examples. During training, the model learns patterns such as grammar, style, structure, color relationships, sound patterns, or coding syntax. During use, it generates an output based on those learned patterns and the prompt it receives.

4.2 Training, Fine-Tuning, and Inference



- **Pretraining:** The model learns broad patterns from a very large dataset.
- **Fine-tuning:** The model is adapted to a specific domain, task, or style.
- **Inference:** The trained model is used to generate a response for a real user prompt.

4.3 Text Generation Intuition

📖 Next-Token Prediction

Large language models often generate text by predicting the most suitable next token repeatedly. Over many steps, this creates a sentence, then a paragraph, then a complete answer.

🔗 Mini Example

Prompt: *“Write one sentence about data science.”*

The model does not retrieve one fixed sentence from memory. Instead, it generates the response token by token based on probabilities learned from training data and the current prompt.

5 Main Families of Generative Models

Model Family	Common Output	Key Idea
Large Language Models (LLMs)	Text, summaries, code, chat responses	Generate text token by token using transformer-based architectures
Diffusion Models	Images, audio, video	Start from noise and gradually refine it into meaningful content
GANs	Images, synthetic samples	A generator competes with a discriminator to create realistic outputs
VAEs	Compressed or synthetic data	Learn a compact latent representation and generate samples from it

5.1 Large Language Models

LLM

A Large Language Model is a generative model trained on large amounts of text. It can answer questions, summarize content, translate language, generate code, and hold conversations.

5.2 Diffusion Models

Diffusion Model

Diffusion models are widely used for image generation. They learn how to reverse noise. During generation, they begin with random noise and gradually transform it into a meaningful image that matches the prompt.

5.3 GANs and VAEs

- **GANs** use two networks: one generates content and the other judges whether it looks real.
- **VAEs** learn a compressed latent space and then generate samples from it.
- Both are important historically, even though modern text generation is dominated by transformer-based models.

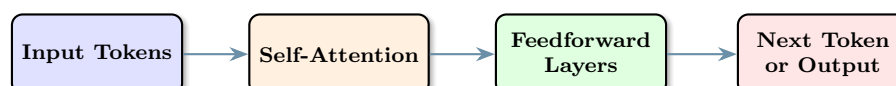
6 Transformers and Attention

Transformer

The transformer is a neural network architecture that became the foundation of modern large language models. It is powerful because it can analyze relationships between different parts of the input efficiently.

Attention

Attention allows the model to focus on the most relevant words, symbols, or regions of the input when generating the next part of the output. This is why transformers are effective for long and complex sequences.



7 Prompting Basics

Prompt

A prompt is the instruction, question, context, or example given to a generative model in order to guide its output.

7.1 What Makes a Good Prompt?

- Be clear about the task.
- Provide enough context.
- State the desired format of the answer.
- Add constraints such as length, tone, or audience.
- Include an example when needed.

Prompt Engineering Example

Weak prompt: *“Explain AI.”*

Better prompt: *“Explain generative AI to first-year BSIT students in five simple bullet points and include one real-life example.”*

7.2 Common Prompting Styles

Prompting Style	Meaning
Zero-shot	Ask the task directly without examples
One-shot	Provide one example before asking for output
Few-shot	Provide a few examples to guide the format or style
Instruction prompting	Tell the model exactly what to do
Role prompting	Ask the model to respond as a tutor, analyst, or assistant

8 Retrieval, Grounding, and RAG

Grounding

Grounding means connecting the model to trusted information sources so its answers are based on real documents, databases, or knowledge rather than only what it learned during training.

RAG — Retrieval-Augmented Generation

RAG combines information retrieval with generation. First, the system retrieves relevant content from a trusted source. Then, the model uses that retrieved content to generate a more accurate answer.

💡 Why RAG Is Useful

RAG is helpful when we need more up-to-date or domain-specific answers, such as university policies, company documents, manuals, or course notes.

9 Applications of Generative AI

- **Text generation:** drafting reports, summaries, emails, and study notes.
- **Conversational systems:** virtual assistants, tutoring systems, and help desks.
- **Code generation:** suggesting functions, debugging help, and documentation.
- **Image generation:** marketing graphics, concept art, and design prototypes.
- **Audio generation:** speech synthesis, dubbing, and voice assistants.
- **Video generation:** short clips, explainers, and animation support.
- **Synthetic data:** generating additional training data when real data is limited or sensitive.

9.1 Examples Relevant to BSIT Data Science

🔗 Classroom-Relevant Use Cases

- Generate study summaries from lecture notes.
- Create chatbot support for campus information.
- Assist with code explanation and documentation.
- Produce draft data reports from structured results.
- Generate synthetic examples for model testing and practice.

10 Limitations and Risks

⚠️ Generative AI Is Powerful but Not Perfect

Generative models can sound confident even when they are wrong. They are useful assistants, but they are not guaranteed truth machines.

Risk	What It Means	Possible Response
Hallucination	The model invents incorrect facts	Verify with trusted sources and grounding
Bias	Outputs may reflect unfair patterns in training data	Review data and test outputs carefully
Privacy issues	Sensitive data may be exposed or mishandled	Avoid sharing private information in prompts
Misinformation	Generated content may spread false claims	Human review before publishing or acting on outputs

Risk	What It Means	Possible Response
Overreliance	Users may trust the tool too much	Keep human judgment in the loop

10.1 Hallucination

Hallucination

In generative AI, hallucination means the model produces content that sounds believable but is false, unsupported, or made up.

10.2 Prompt Injection and Safety

Prompt Injection

Prompt injection happens when malicious or misleading instructions try to manipulate a model or override its intended behavior. This matters when generative AI systems are connected to tools, files, or external data sources.

11 Evaluating Generative AI Systems

- **Relevance:** Does the answer address the prompt?
- **Accuracy:** Is the content factually correct?
- **Clarity:** Is the response easy to understand?
- **Groundedness:** Is the answer supported by source material?
- **Safety:** Does the output avoid harmful, biased, or toxic content?
- **Latency:** How quickly does the system respond?

Human Evaluation Still Matters

Unlike simple classification tasks, generative AI often needs human judgment because quality depends on context, usefulness, tone, and correctness.

12 Best Practices for Students and Developers

- Start with a clear use case.
- Use trusted data sources when accuracy matters.
- Write specific prompts instead of vague prompts.
- Validate important outputs before using them.
- Protect personal, confidential, and copyrighted data.
- Keep a human reviewer in the workflow.
- Compare model outputs with baseline methods when possible.
- Document how the system is used and what its limits are.

13 Mini Case Study for BSIT

Campus Helpdesk Assistant

Suppose a BSIT department wants to build a chatbot that answers common student questions about schedules, enrollment steps, laboratory rules, and course requirements.

Possible design:

1. Collect official FAQ documents and campus announcements.
2. Store them in a searchable knowledge base.
3. Use a generative model with RAG so answers are based on official documents.
4. Add safety rules so the system does not invent policies.
5. Let staff review important updates and monitor failures.

Responsible Use

For academic or administrative systems, generated answers should support staff and students, not replace official human approval where accuracy is critical.

14 A Small API-Style Example

```
prompt = "Summarize the topic Generative AI for BSIT students in 5  
bullet points."  
  
response = model.generate(  
    prompt=prompt,  
    temperature=0.4,  
    max_tokens=200  
)  
  
print(response)
```

What This Example Shows

The prompt gives the task, the model generates text, and settings such as temperature and maximum tokens influence the style and length of the response.

15 Summary

- Generative AI creates new content such as text, images, code, and audio.
- It differs from predictive AI because its main goal is content creation, not only classification or forecasting.
- Core concepts include prompts, tokens, embeddings, context windows, inference, and fine-tuning.
- Important model families include LLMs, diffusion models, GANs, and VAEs.
- Transformers and attention are central to modern text-based generative AI.

- RAG helps improve accuracy by connecting the model to trusted external information.
- Generative AI has major benefits, but it also has risks such as hallucination, bias, and privacy problems.

16 Review Questions

1. What is generative AI, and how is it different from predictive AI?
2. What is a prompt?
3. What is a token, and why is it important in large language models?
4. What is meant by a context window?
5. What is the difference between pretraining, fine-tuning, and inference?
6. How do LLMs differ from diffusion models?
7. What does attention do in a transformer?
8. What is hallucination in generative AI?
9. Why is RAG useful in real applications?
10. What ethical and practical risks should users consider when using generative AI?